



equicenta[®] CTM

Clinical Evidence Booklet

Regenerative musculoskeletal tissue support for horses — peer-reviewed in *Frontiers in Veterinary Science*, 2026.

“Horse First, Sport Second.”

Equine Performance Labs’ guiding principle for equicenta[®] CTM development.

PRIMARY PUBLICATION

Bertone AL, Davern A, Sutter WW, et al. Cryopreserved umbilical allograft (equicenta[®] CTM) improves clinical outcomes compared with clinician-selected standard care in performance horses: a pragmatic prospective controlled trial. *Front. Vet. Sci.* 2026. doi:10.3389/fvets.2026.1833933



Study at a glance

A pragmatic prospective controlled trial in performance horses — peer-reviewed in *Frontiers in Veterinary Science*, 2026.

Peer-reviewed clinical trial data

Study registry: IACUC ETCR-2024-0329 · AVMA VCT #26005934

By the numbers

138

HORSES

performance cohort

195

CONDITIONS

joint · ligament · tendon

8

PRACTICES

11 veterinarians

26

WEEKS

performance follow-up

1.9:1

ALLOCATION

eCTM : SOC

Abstract (plain language)

In a multicenter controlled trial, a single injection of equicenta[®] CTM — a cryopreserved equine umbilical cord tissue allograft — was compared with clinician-selected standard-of-care biologics. Horses receiving eCTM showed greater improvement in pain, swelling, and lameness through 12 weeks, higher odds of returning to prior performance at 26 weeks, and a favorable injection-reaction profile. Results are peer-reviewed; the product remains investigational.

Return to — or exceeding — prior performance through 26 weeks (eCTM vs SOC)

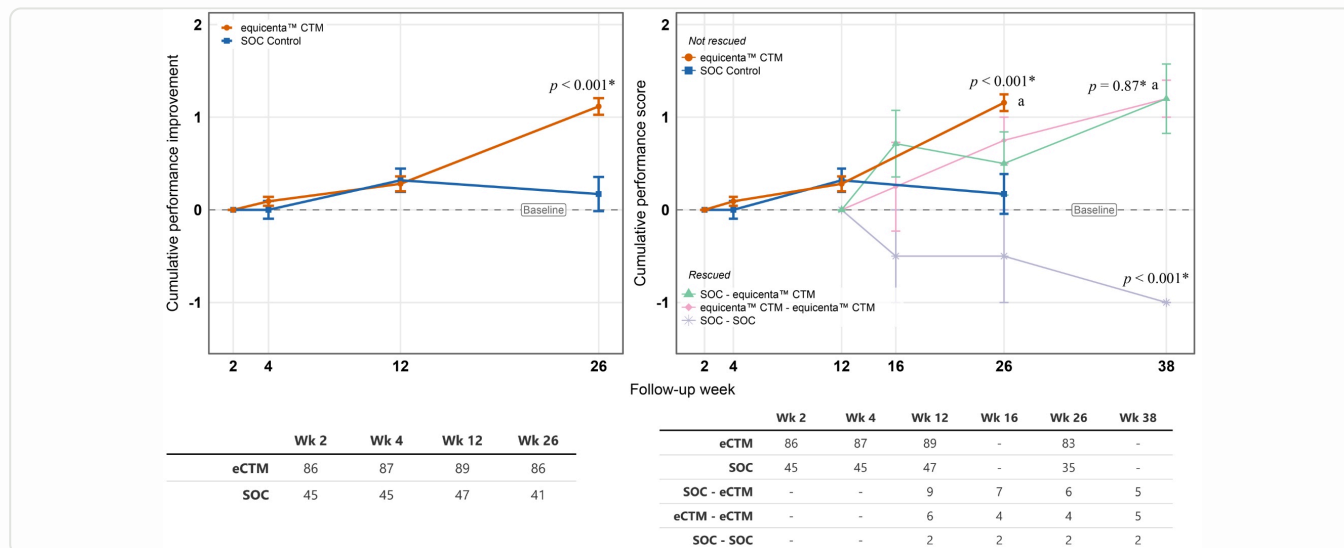


Figure 3. Cumulative performance improvement over follow-up: equicenta[®] CTM (orange) vs clinician-selected SOC control (blue). ITT 26-week effect size Cohen's $d = 2.23$, $p < 0.001$. Error bars = SEM.

KEY RESULT

26-week performance effect size vs. reference thresholds — small (0.2), medium (0.5), large (0.8); observed $d = 2.23$, far beyond the large-effect threshold.

SECTION 01 · PRODUCT & INDICATION CONTEXT

equicenta[®] CTM — cryopreserved umbilical cord tissue allograft

A ready-to-use orthobiologic studied for joint, tendon, and ligament conditions in performance horses.

What it is

equicenta[®] CTM is a cryopreserved equine umbilical cord–derived connective tissue matrix (CTM) allograft for local administration. In the published trial, every horse in the eCTM arm received a single injection as studied — without a multi-step autologous processing series in that arm. The comparator was clinician-selected standard of care (SOC) biologics used in real-world practice — a pragmatic design intended to reflect field decision-making rather than an artificial single-comparator lab model.

Product specification

TISSUE SOURCE	Equine umbilical cord–derived connective tissue matrix (allograft)
PRESERVATION	Cryopreserved; ready-to-use, no processing series required at time of injection
REGIMEN STUDIED	Single local injection per treated condition (as administered in the trial)
COMPARATOR	Clinician-selected standard-of-care biologic (real-world practice choice)
CONDITIONS STUDIED	Joint, tendon, and ligament conditions — 195 conditions across 138 horses
REGULATORY STATUS	Investigational — not approved or cleared by U.S. FDA Center for Veterinary Medicine

KEY TAKEAWAY

A single studied injection, compared in a peer-reviewed trial against the standard-of-care biologics already in use. Findings are intended to inform discussion between veterinarian and owner — not to substitute for clinical judgment.

SCOPE OF CLAIMS

equicenta[®] CTM has not been shown to cure disease, and absolute safety has not been established. Exploratory analyses in this booklet are identified as exploratory throughout.

SECTION 02 · STUDY DESIGN

First controlled clinical trial of a tissue allograft in horses

Prospective, multicenter, quasi-randomized, open-label, controlled pragmatic trial.

138 HORSES enrolled	195 CONDITIONS treated	8 U.S. PRACTICES 11 veterinarians	1.9:1 ECTM : SOC 91 : 47	100% WK12 COMPLETE 138/138	98% WK26 RETAINED 135/138
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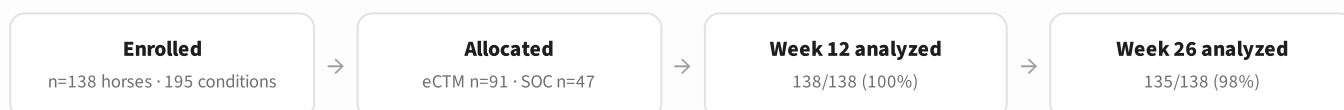
METHODS ESSENTIALS

- Clinician-selected SOC biologic comparator (real-world care).
- Quasi-randomized within practice to a pre-specified allocation target.
- Primary clinical window: Weeks 0, 2, 4, 12 (lameness, pain, swelling).
- Performance and owner-reported outcomes through Week 26.
- Blinded secondary assessments: Tissue Remodeling Score ultrasound; Sleip kinematics.
- CONSORT-guided reporting; third-party biostatistics.
- IACUC ETCR-2024-0329 · AVMA VCT #26005934.

WHY THE DESIGN MATTERS

Case series can show that some horses improved. A controlled trial asks a harder question: did this product outperform the biologics already used in practice, under the same follow-up rules?

Near-complete follow-up (100% at the Week-12 primary clinical endpoint; 98% at Week 26) reduces the risk that results reflect only the easiest cases to keep in the study.



HEADLINE RESULT (PUBLICATION)

eCTM outperformed SOC across clinical outcomes by 12 weeks ($p < 0.001$) and showed a large performance advantage at 26 weeks (Cohen's $d = 2.23$, $p < 0.001$), with higher odds of return to prior performance (OR 2.72, $p < .006$).

Source: Bertone AL, Davern A, Sutter WW, et al. Front. Vet. Sci. 2026. Results §3.1; Methods §2.1–2.9; Figure 1. doi:10.3389/fvets.2026.1833933

SECTION 03 · PUBLICATION CONCLUSIONS

Overarching conclusion from the peer-reviewed paper

Quoted directly from the peer-reviewed publication.

“*In real-world lame performance horses, a single dose of equicenta® CTM demonstrated superior safety and effectiveness over standard-of-care biologic therapies through 26 weeks, supporting its use as a consistent, single-dose alternative to standard-of-care for selected musculoskeletal conditions as a cryopreserved, ready-to-use orthobiologic for joint, ligament, and tendon conditions. To the authors' knowledge, this represents the first controlled evidence for an allograft musculoskeletal connective tissue therapy in animals.*”

Five evidence pillars

01 Clinical	Greater cumulative improvement in pain, swelling, and lameness through Week 12 ($p < 0.001$). Separation by Week 2 (pain/swelling) and Week 4 (lameness).
02 Performance	Large ITT performance effect at 26 weeks ($d = 2.23$, $p < 0.001$). Return to prior performance OR 2.72 ($p < .006$).
03 Safety	Lower injection-reaction odds vs SOC (vet 4.8:1; central 5.9:1). Per-injection reactions 3.4% eCTM vs up to 10.5% SOC. No eCTM product-related SAEs reported.
04 Owner view	Week-26 owner scores favored eCTM for body condition, coat, overall health, and quality-of-life index.
05 Structure*	Exploratory ultrasound: Responding tissue-remodeling score 36% eCTM vs 0% SOC ligaments ($p = 0.013$).

*EXPLORATORY QUALIFIER

Imaging and some subgroup contrasts are exploratory analyses. Controlled clinical outcomes, performance odds, and safety are the primary, pre-specified evidence in this study.

Source: Bertone AL, Davern A, Sutter WW, et al. Front. Vet. Sci. 2026. Discussion; Results §3.4. doi:10.3389/fvets.2026.1833933

SECTION 04 · CLINICAL EFFECTIVENESS

Earlier separation — sustained through the Week-12 clinical window

Lameness, pain, and swelling versus clinician-selected standard of care.

$p < 0.001$ ALL CLINICAL OUTCOMES through Week 12	Wk 2 PAIN & SWELLING first separation	Wk 4 LAMENESS significant	3:1 PAIN-FREE ODDS Week 12 · $p < 0.001$	>1.6:1 SOUNDNESS ODDS directional · $p < 0.06$
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- Faster and greater cumulative improvement.** eCTM produced greater cumulative improvement in pain, swelling, and lameness than SOC. Significance by Week 2 for pain ($p < 0.02$) and swelling ($p < 0.02$), by Week 4 for lameness ($p < 0.01$), sustained through 12 weeks ($p < 0.001$ all).
- Complete resolution odds.** Odds of score-zero at Week 12 favored eCTM: pain-free 3:1 ($p < 0.001$); soundness (lameness 0) >1.6:1 (near-threshold — communicate directionally).
- Clinical window boundary.** Clinical scores are reported through Week 12. Performance is a separate 26-week endpoint and must not be mixed into clinical score trajectories.

STATISTICAL SIGNIFICANCE BY MEASURE/TIMEPOINT

Pain (Week 2)		$p < 0.02$
Swelling (Week 2)		$p < 0.02$
Lameness (Week 4)		$p < 0.01$
All measures (Wk 12)		$p < 0.001$

Bar length = $-\log_{10}(p)$; longer bar indicates stronger statistical significance.

IN PLAIN TERMS

Horses treated with eCTM typically started looking more comfortable within the first recheck window and remained ahead on scored clinical measures through three months.

WHAT “STANDARD OF CARE” MEANT HERE

The comparator arm was not a placebo. Each SOC joint received a clinician-selected, **non-steroidal** therapy — most often autologous blood-derived orthobiologics, and also amnion allograft, hyaluronic acid, or polyacrylamide gels — so eCTM was measured against real-world best practice, not against no treatment.

HEAD-TO-HEAD VS. HEMODERIVATIVES

A dedicated head-to-head contrast against the largest SOC class — blood-derived orthobiologics (APS, PRP, ACS, PPP) — with the published Figure 8 trajectories appears in **Section 09**.

Source: Bertone AL, Davern A, Sutter WW, et al. Front. Vet. Sci. 2026. Results §3.4.1.1. doi:10.3389/fvets.2026.1833933

SECTION 05 · PERFORMANCE & RETURN-TO-WORK

Return to — or exceeding — prior performance at 26 weeks

A pre-specified performance endpoint from the published trial.

OR 2.72 RETURN-TO-PERFORMANCE $p < .006$	d = 2.23 EFFECT SIZE (ITT) $p < 0.001$	52.9% ECTM SCORE-4 45/85 within-arm	29.3% SOC SCORE-4 12/41 within-arm
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WITHIN-ARM SCORE-4 RATES (RETURN TO OR EXCEED PRIOR PERFORMANCE)

equicenta® CTM (single injection)	45/85
Standard of care (clinician-selected)	12/41

ADDITIONAL PERFORMANCE FINDINGS

- Mean performance was higher for eCTM at every evaluation point in the study window.
- ITT performance advantage at 26 weeks was large (Cohen's $d = 2.23$, $p < 0.001$).
- Advantage preserved in intention-to-treat and per-protocol analyses.

IN PLAIN TERMS

Horses that received eCTM had nearly three times the odds of getting back to their prior performance level compared with horses on standard biologics.

READING THIS STATISTIC CORRECTLY

Elsewhere, the trial's Score-4 outcomes are sometimes summarized as 79% eCTM / 21% SOC — this describes the composition of the 57 horses who achieved Score-4 (45 eCTM / 12 SOC), not a within-arm success rate. The within-arm rates are 52.9% (eCTM) vs 29.3% (SOC), corresponding to OR 2.72.

Performance trajectory and rescue crossover (eCTM vs SOC)

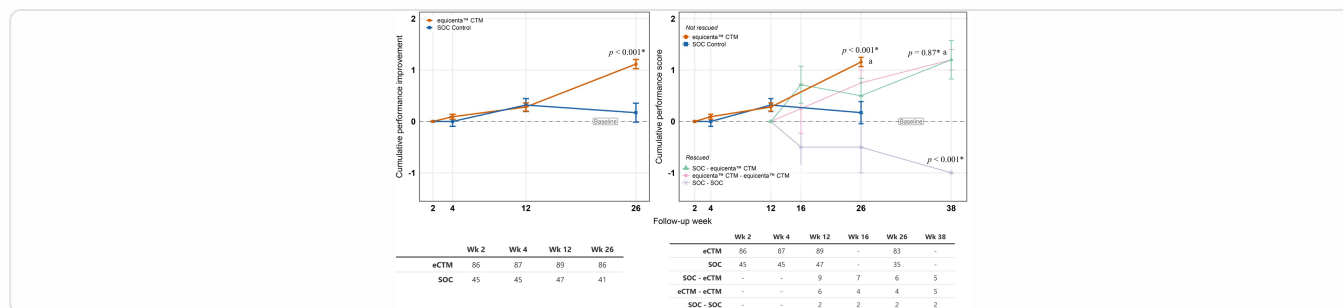


Figure 3. Left: cumulative performance improvement, eCTM (orange) vs SOC (blue), through Week 26 (ITT $d = 2.23$, $p < 0.001$). Right: rescue crossover (exploratory/secondary) — SOC→eCTM recovered to levels indistinguishable from first-line eCTM ($p = 0.87$); SOC→SOC declined ($p < 0.001$). Error bars = SEM.

Source: Bertone AL, Davern A, Sutter WW, et al. Front. Vet. Sci. 2026. Results §3.4.2.1; Figure 3. doi:10.3389/fvets.2026.1833933

SECTION 06 · SAFETY

Injection reactions and injection-site inflammation

Head-to-head observation versus clinician-selected standard of care in the same trial.

3.4% ECTM REACTIONS 5/148 injections	10.5% SOC REACTIONS 8/76 · central review	4.8:1 VET-REPORTED ODDS SOC higher · $p < 0.04$	5.9:1 CENTRAL REVIEW ODDS SOC higher · $p = 0.002$	0/148 ISI SCORE >0 eCTM injections
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ECTM ARM

- 5 reactions across 148 injections (3.4% per-injection).
- 0/148 injections with injection-site inflammation (0–4 scale) at weeks 2, 4, and 12.
- 0/148 infections; no vet-reported product-related serious adverse events.
- Observed reactions were mild, self-limited, and early.

SOC ARM (CONTEXT)

- 8 reactions / 76 injections under central review (10.5%).
- 2/76 serious intra-articular reactions in SOC; none with eCTM.
- Advantage held on vet report and blinded central adjudication.
- PAAG contributed to excess SOC joint reactions; excluding PAAG, SOC odds remained higher (3.4:1, $p < 0.03$).

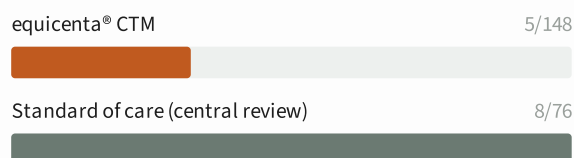
IN PLAIN TERMS

Standard biologics in this study were associated with several times more injection reactions. No serious product-related events were attributed to eCTM.

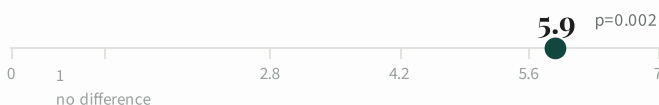
INVESTIGATIONAL STATUS

equicenta® CTM is investigational. No side effects or absolute safety have been established, and individual results may vary.

PER-INJECTION REACTION RATE



SOC-VS-ECTM REACTION ODDS (CENTRAL REVIEW)



Source: Bertone AL, Davern A, Sutter WW, et al. Front. Vet. Sci. 2026. Results §3.4.1.2. doi:10.3389/fvets.2026.1833933

SECTION 07 · OWNER OUTCOMES & STRUCTURAL SIGNAL

What owners reported — and what imaging explored

Secondary owner endpoints at Week 26; exploratory tissue remodeling on ultrasound.

A. OWNER-REPORTED OUTCOMES (WEEK 26)

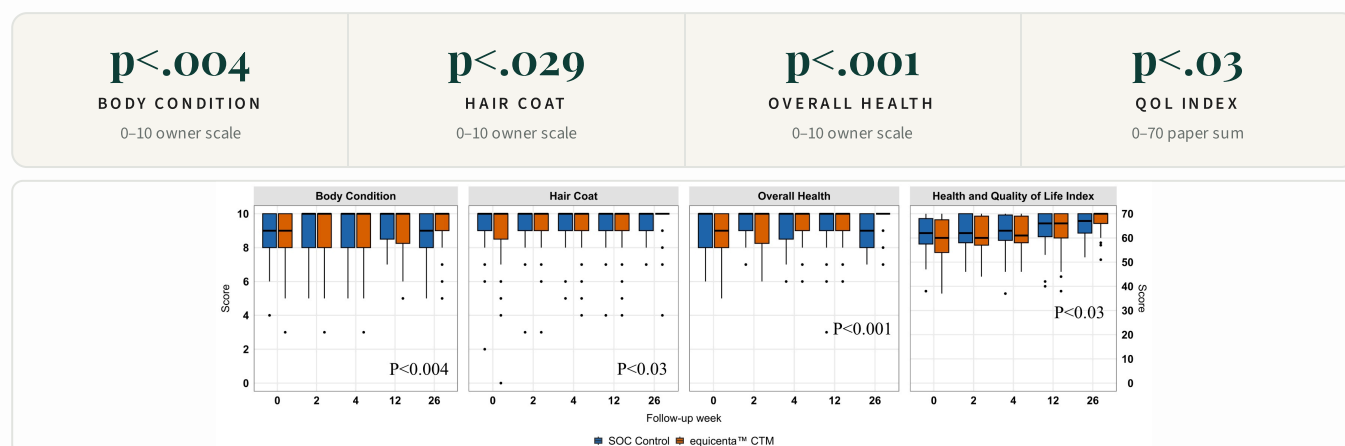
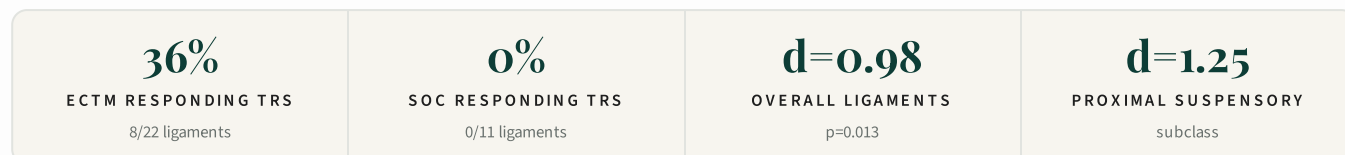


Figure 4. Owner-reported secondary outcomes at Week 26 (eCTM orange vs SOC blue). Direction of improvement aligned with clinician-scored outcomes. QOL index uses a 0–70 scale, reported as such (not converted to 0–10).

B. STRUCTURAL HEALING — EXPLORATORY TRS



Final tissue remodeling score (TRS)

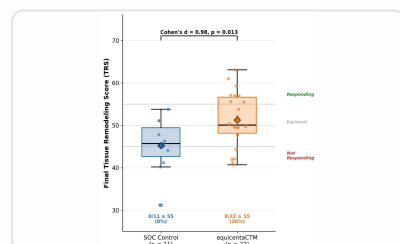
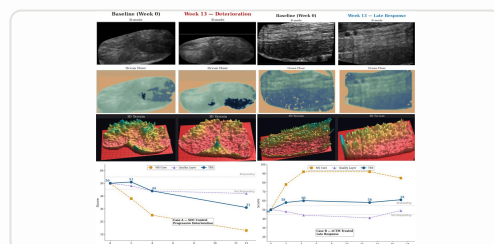


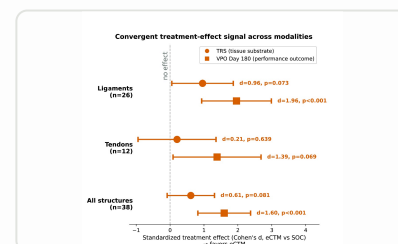
Figure 5. Final TRS by arm: eCTM (orange) vs SOC (blue). Cohen's d=0.98, p=0.013; responding (≥ 55) 8/22 eCTM vs 0/11 SOC. Exploratory.

TRS ultrasound case examples (baseline > Week 13)



Suppl. Fig. S1. Case A (SOC): deterioration, TRS 50→31. Case B (eCTM): late response, TRS 50→61 (crossing responding threshold). Exploratory.

Convergent treatment-effect signal



Suppl. Fig. S2. TRS vs performance effect sizes across ligaments, tendons, all structures. Exploratory; tendon estimate small (n=13).

Source: Bertone AL, Davern A, Sutter WW, et al. Front. Vet. Sci. 2026. \$3.4.2.2 Fig 4; \$3.4.2.4 Fig 5; Suppl. Figs S1–S2. doi:10.3389/fvets.2026.1833933

SECTION 08 · RIGOR, FOLLOW-UP & ONSET

Study integrity and when improvement separates

Design safeguards documented in the peer-reviewed publication.

A. RIGOR & FOLLOW-UP

138/138 WEEK 12 100% primary	135/138 WEEK 26 98% retained	Blinded SECONDARIES US · kinematics · stats	CONSORT REPORTING pre-specified
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- Independent third-party biostatistics; analysts blinded where specified.
- Standardized AAEP lameness and validated 0–4 pain/swelling scores.
- Bias safeguards: eligibility, consecutive enrollment, stratified allocation, consensus training.
- Study rigor is documented ahead of the efficacy findings in the manuscript.
- The DOI and trial IDs are provided for independent verification.
- Complete follow-up supports both intention-to-treat and per-protocol interpretation.

B. ONSET TIMELINE (CLINICAL WINDOW)



IN PLAIN TERMS

Improvement is not only a six-month story. Clinical separation begins at the early rechecks and is measured rigorously through Week 12; return-to-work is then confirmed at Week 26.

CONSECUTIVE Enrollment, not selective	STRATIFIED Allocation by practice	INDEPENDENT Third-party biostatistics	CONSENSUS Scorer training
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Source: Bertone AL, Davern A, Sutter WW, et al. Front. Vet. Sci. 2026. \$3.1; Fig 1; \$3.4.1.1 Fig 2. doi:10.3389/fvets.2026.1833933

SECTION 09 · HEAD-TO-HEAD VS. HEMODERIVATIVES

Tested against the prevailing biologic paradigm

A pre-planned contrast of eCTM against pooled autologous blood-derived products — the largest treatment class in the standard-of-care arm.

63.2% OF THE SOC ARM were hemoderivatives	n=43 HEMODERIVATIVE joints (SOC)	p<0.04 LAMENESS Weeks 4 & 12	p<0.003 PAIN on flexion	p<0.001 SWELLING effusion
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- The comparator that matters.** Autologous blood-derived orthobiologics — APS, PRP, ACS, and PPP — are the dominant intra-articular biologics in equine practice. In this pragmatic trial they formed the largest single subgroup of the clinician-selected SOC arm (n=43; 63.2%), so this contrast pits eCTM against the prevailing standard.
- eCTM improved more, across timepoints.** In the pooled hemoderivative contrast, eCTM showed greater cumulative improvement at multiple timepoints (range p<0.04 to p<0.001), most clearly on lameness (Weeks 4 & 12, p<0.04), pain (p<0.003), and swelling (p<0.001).
- Single dose vs. the class.** Each eCTM joint received one 1.5 mL intra-articular injection of cryopreserved umbilical-derived protein/microparticle suspension — compared against the range of blood-derived products chosen by experienced clinicians in real-world practice.

HOW TO READ THIS CONTRAST

This is a **pre-planned (predefined) exploratory subgroup analysis**, one of two clinically sensible SOC contrasts identified by the authors. It uses the pooled standard-of-care variance estimate. Report it as supportive, class-level evidence — not as a primary registrational endpoint.

IN PLAIN TERMS

Against the blood-derived products that make up most of everyday joint therapy, a single off-the-shelf eCTM injection produced greater improvement in lameness, pain, and swelling through the 12-week clinical window.

Cumulative improvement from baseline — eCTM vs. pooled hemoderivatives (lameness, pain, swelling)

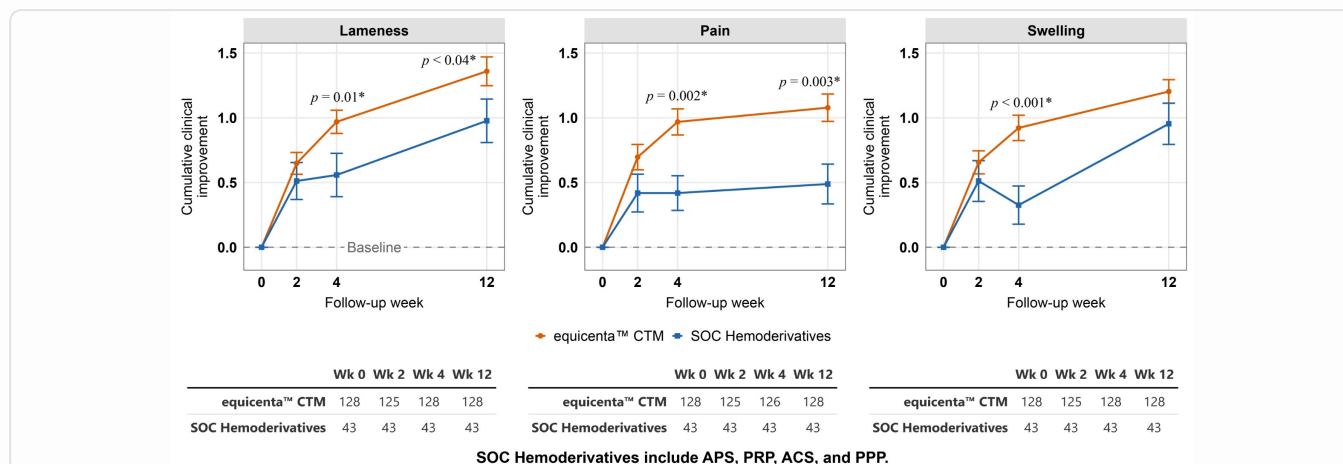


Figure 8. Sub-class analysis for hemoderivatives. Cumulative clinical improvement trajectories from baseline (mean ± SEM) were greater for eCTM (orange) than the SOC subclass of hemoderivatives (blue) at most timepoints over 12 weeks — particularly lameness (Weeks 4 & 12; p<0.04) and pain (p<0.003). Hemoderivatives = APS, PRP, ACS, and PPP. Predefined exploratory subgroup analysis.

Source: Bertone AL, Davern A, Sutter WW, et al. Front. Vet. Sci. 2026. Methods §2.7; Results §3.4 Figure 8; Supplementary Figure S3. doi:10.3389/fvets.2026.1833933

SECTION 09 (CONT.) · HEAD-TO-HEAD VS. HEMODERIVATIVES — MECHANISM & PRACTICE

Why the difference is biologically plausible

Duration of effect, injection burden, and safety — the practice case behind the numbers.

DURATION OF PRIOR EVIDENCE

Controlled evidence for blood-derived orthobiologics extends only to about **2 weeks** (one controlled study of APS), with longer-term outcomes unknown. Their soluble proteins are expected to be rapidly absorbed or degraded, so benefit is anticipated to be short-lived.

ECTM MECHANISM

Umbilical allograft (eCTM) delivers fetal anabolic and reparative proteins, in part as a biodegradable connective tissue that **releases proteins over time** — a plausible mechanism for the more sustained benefit observed relative to the hemoderivative class.

INJECTION BURDEN

Multiple-injection protocols for ACS and PRP are widely used, yet three intra-articular ACS injections did not demonstrate clinical benefit in a controlled study. eCTM was delivered as a single injection (see re-injection rates below).

SAFETY CONSIDERATION

Post-injection infection remains the most frequently reported serious adverse event for PRP and other hemoderivatives. Because any repeated joint injection may raise infection risk, the authors note that justifying multiple injections **should rest on efficacy data or be balanced against options requiring fewer administrations.**

PRACTICE IMPLICATION

Hemoderivatives require blood draw, chairside processing, and often multiple visits. A single, off-the-shelf allograft that outperformed this class on blinded lameness and pain endpoints has meaningful workflow implications.

ADDITIONAL INJECTIONS BY ORIGIN (TABLE 5)



SOC-assigned horses generated **significantly more** additional injections (20 vs. 9; $p < 0.03$), driven largely by protocol-prescribed repeat doses; rescue-injection counts alone did not differ between arms ($p > 0.09$).

APS-SPECIFIC CONTRAST (EXPLORATORY · POST HOC)

Why APS

APS was the single most-used hemoderivative ($n=22$; 51% of hemoderivatives) with prior supportive evidence in equine joint osteoarthritis — so it was chosen for a dedicated contrast.

What was seen

eCTM remained associated with greater improvement at multiple timepoints, primarily in lameness and pain (Supplementary Figure S3).

How to weight it

This APS contrast was **not pre-specified** and was limited by small sample size; treat it as hypothesis-generating, not confirmatory.

EVIDENTIARY NOTE

The pooled-hemoderivative contrast is a predefined exploratory subgroup analysis; the APS-specific contrast is post hoc. Neither is a primary registrational endpoint. equicenta® CTM remains investigational and is not approved or cleared by the U.S. FDA Center for Veterinary Medicine.

Source: Bertone AL, Davern A, Sutter WW, et al. Front. Vet. Sci. 2026. Discussion §4.1–4.3; Results §3.4; Supplementary Figure S3. doi:10.3389/fvets.2026.1833933

SECTION 10 · OBJECTIVE GAIT & COMPARATIVE CONTEXT

Blinded kinematics and exploratory class comparisons

Supportive analyses, reported at their appropriate evidentiary tier.

A. SLEIP KINEMATICS (EXPLORATORY)

$p < .003$ PUSH-OFF Week 2	$p < .05$ IMPACT Week 4	$p < .01$ PUSH-OFF Week 12	Blinded ANALYSTS to treatment
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- Reported p-values cover Weeks 2, 4, and 12 only; Week 26 kinematic data were not reported.
- Sensor-based kinematics support the direction of clinical findings and do not replace the primary clinical endpoints.

Cumulative gait asymmetry from baseline (lower = improvement toward symmetry)

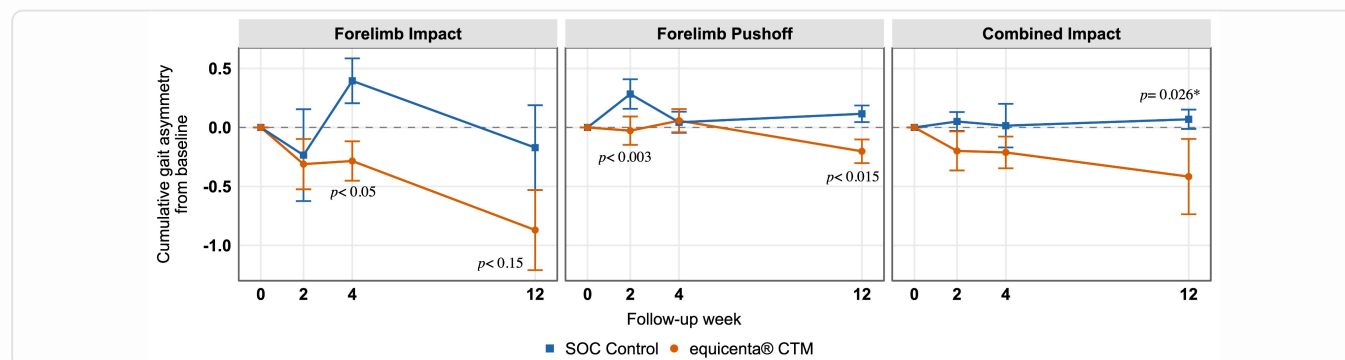


Figure 6. Sleip inertial-sensor kinematics: forelimb impact, forelimb push-off, combined impact. eCTM (orange) vs SOC (blue). Blinded analysts. Exploratory supportive endpoint. Error bars = SEM.

B. RESCUE CROSSOVER (EXPLORATORY / SECONDARY)

Rescue criterion

19 horses met the pre-defined rescue criterion for inadequate response; 17 were analyzable after crossover.

SOC→eCTM recovery

Horses crossed from SOC to eCTM recovered to levels statistically indistinguishable from first-line eCTM ($p=0.87$).

Hemoderivative contrast

The full head-to-head vs. blood-derived products (APS, PRP, ACS, PPP), with Figure 8, is presented in Section 09.

EVIDENTIARY NOTE

These comparative and rescue analyses are exploratory or post-hoc unless stated otherwise.

Source: Bertone AL, Davern A, Sutter WW, et al. Front. Vet. Sci. 2026. §3.4.2.5 Fig 6; §3.4.2.7; rescue §3.4.2.1. doi:10.3389/fvets.2026.1833933

SECTION 11 · DISCUSSION — CLINICAL RELEVANCE

Why the effect is durable — and what it means at chairside

Dr. Bertone and co-investigators interpret the mechanism, timing, and structural signals behind the clinical separation.

Earlier recovery, longer benefit

eCTM-treated conditions showed **earlier divergence and larger cumulative improvements** in lameness, pain, and swelling than SOC, with differences emerging by 2–4 weeks and persisting for at least 12 weeks.

Hemoderivative benefit is expected to be short-lived — soluble proteins are rapidly absorbed or degraded, and one controlled study showed APS benefit out to only 2 weeks. Umbilical allograft instead delivers fetal anabolic and reparative proteins as a biodegradable connective tissue, **releasing proteins over time** — a plausible mechanism for the sustained benefit observed.

Consistent across tissue types

The uniformity of response across joints, tendons, and ligaments was anticipated: many biological processes are shared among musculoskeletal connective tissues, and this is the established rationale for using biologics in these same tissues.

CHAIRSIDE IMPLICATION

Between 12 and 26 weeks, SOC performance gains plateaued or declined — particularly in soft-tissue conditions — whereas eCTM horses continued to improve, suggesting eCTM may help sustain functional recovery.

A single injection vs. repeat protocols

Multiple-injection ACS and PRP protocols are widely used, so SOC received significantly more re-injections than the single eCTM injection. Yet three intra-articular ACS injections did not show clinical benefit in a controlled equine study, and a joint infection occurred after a third injection.

Post-injection infection remains the most frequent serious adverse event for hemoderivatives. Because any repeated joint injection may increase infection risk, the authors note the justification for multiple injections **should rest on efficacy data or be balanced against options requiring fewer administrations.**

Joint pain and performance move together

SOC-treated joints had transient performance improvement not matched by sustained joint-pain relief, then declined toward baseline by 26 weeks. eCTM-treated joints showed **sustained joint-pain improvement with continued performance gains** through 26 weeks.

HYPOTHESIS-GENERATING

As sub-cohort analyses, these suggest eCTM may provide more durable modulation of intra-articular synovial pain than some existing biologics; definitive mechanistic studies would be needed to test this hypothesis.

Source: Bertone AL, Davern A, Sutter WW, et al. Front. Vet. Sci. 2026. \$4.1 Clinical effectiveness and durability of response. doi:10.3389/fvets.2026.1833933

SECTION 12 · DISCUSSION — STRUCTURE, SAFETY & DESIGN

Structural signals, a favorable safety profile, and a real-world design

Supportive structural and kinematic evidence, owner-reported outcomes, and the pragmatic trial design — with limitations stated plainly.

A. STRUCTURAL REMODELING & QUANTITATIVE VALIDATION (EXPLORATORY)

Quantitative ultrasound-based **TRS** provided complementary structural evidence of tissue remodeling — particularly in ligament lesions — that may coincide with later performance improvement. Segmentation reproducibility was high, with a **cohort mean Dice coefficient of 0.931**, above published strong intra-rater values.

Tendon TRS correlated significantly with 12-week limb-level lameness and was positively associated with 26-week performance. These findings **do not constitute full validation** but provide exploratory hypothesis generation to justify further work.

Kinematic gait analysis by **blinded analysts** supported the clinical findings, particularly in forelimb lameness: forelimb impact and push-off asymmetry improved more in eCTM horses at several time points. Hindlimb and combined-limb data were highly variable.

EVIDENTIARY TIER

TRS is a novel, not-yet-fully-validated biomarker used here as an exploratory endpoint; kinematics are a supportive endpoint. Both reinforce — but do not replace — the primary clinical outcomes.

B. SAFETY & OWNER-REPORTED OUTCOMES

eCTM was **well tolerated**; product-related adverse events were infrequent, self-limited, and early. SOC treatments were associated with a higher frequency of product-related events overall and in joints specifically, including a small number of serious events. **Two serious joint reactions occurred after polyacrylamide**, one persisting at follow-up. Central review that reclassified events as possibly product-related did not change the safety profile favoring eCTM.

At 26 weeks, owners of eCTM horses reported **higher scores for body condition, coat quality, overall health, and composite quality of life** than SOC owners — suggesting the product will be well received and may reflect subclinical measures of health.

C. PRAGMATIC DESIGN & STATED LIMITATIONS

The trial intentionally reflected real-world practice — 8 high-volume practices, a spectrum of joint and soft-tissue conditions, and an SOC arm spanning treatments in current clinical use — enhancing external validity. The authors state plainly that the study was **open-label and unblinded within practices** (SOC required visible blood collection), mitigated by quasi-randomized allocation, prespecified definitions, standardized scoring, and objective TRS/kinematic measures. Subgroup analyses are **hypothesis-generating and some underpowered**; the trial was powered for the overall eCTM-vs-SOC comparison.

Source: Bertone AL, Davern A, Sutter WW, et al. Front. Vet. Sci. 2026. §4.2 Structural remodeling; §4.3 Pragmatic design; §4.4 Safety and owner-reported outcomes; §4.5 Limitations. doi:10.3389/fvets.2026.1833933

SECTION 13 · CONCLUSION — CLINICAL IMPLICATIONS

A consistent, single-dose option — where prior biologics fell short

The investigators' clinical implications and summary, in their own framing.

“*In this pragmatic setting, a single off-the-shelf injection of eCTM was associated with earlier, greater, and more durable improvements in clinical and performance outcomes than clinician-selected non-steroidal SOC. Overall, these positive findings suggest eCTM may represent a consistent, single-dose alternative to clinician SOC approaches for selected musculoskeletal conditions in performance horses.*”

What the evidence supports

- 1 **Earlier & greater improvement.** In lameness, pain, and swelling versus standard care currently used in practice.
- 2 **Sustained performance.** Gains maintained through the 26-week endpoint.
- 3 **Favorable safety.** Infrequent, self-limited, early product-related events.
- 4 **Supportive structure & gait.** Consistent exploratory TRS and blinded kinematic signals.
- 5 **Practical delivery.** Consistent composition, no patient-side processing, single injection.

Where it fits in practice

The authors specifically support consideration of eCTM **in horses with performance-limiting lameness where previous biologic treatments have provided only transient benefit**, as for the horses in this study.

Future directions

Planned work includes stratified cohorts, optimal dosing, and re-treatment strategies, plus mechanistic studies integrating imaging, biomarker, and clinical data to clarify how eCTM influences tissue remodeling and how best to select candidates most likely to benefit.

STUDY CONTEXT

This multicenter, quasi-randomized, open-label controlled trial with long-term follow-up on over a hundred horses in real-world conditions helps fill a knowledge gap; fully blinded RCTs have not been done in horses and carry their own trade-offs in generalizability.

AUTHORS' INDEPENDENCE

All authors were third-party independent collaborators and not employees of EPL. Dr. Alicia Bertone served as an independent research consultant, Project Monitor, and Multi-center Director for the study.

Source: Bertone AL, Davern A, Sutter WW, et al. Front. Vet. Sci. 2026. \$4.6 Clinical implications and future directions; \$5 Summary and conclusion; Author's note. Product investigational — not FDA CVM approved. doi:10.3389/fvets.2026.1833933

Evidence from the peer-reviewed record.

This booklet summarizes the peer-reviewed Bertone et al. 2026 controlled trial of equicenta® CTM, pairing manuscript-aligned statistics with the study's own published figures.

PUBLICATION	Bertone AL, Davern A, Sutter WW, et al. Front. Vet. Sci. 2026
DOI	10.3389/fvets.2026.1833933
JOURNAL	Frontiers in Veterinary Science · Regenerative & Sports Medicine
TRIAL IDS	IACUC ETCR-2024-0329 · AVMA VCT #26005934
PRODUCT STATUS	Investigational — not FDA CVM approved

equicenta® CTM

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Read the paper: [doi:10.3389/fvets.2026.1833933](https://doi.org/10.3389/fvets.2026.1833933)

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IMPORTANT INFORMATION

Important information: equicenta® CTM is under investigation. It has not been approved by the U.S. FDA Center for Veterinary Medicine, and the safety and effectiveness of the product for the uses described have not been established. Results are from a single prospective controlled clinical study in performance horses; individual outcomes may vary. Not for human use. Not for use in food-producing animals. Source: Bertone AL, Davern A, Sutter WW, et al. Cryopreserved umbilical allograft (equicenta® CTM) improves clinical outcomes compared with clinician-selected standard care in performance horses: a pragmatic prospective controlled trial. Front. Vet. Sci. 2026. doi:10.3389/fvets.2026.1833933. IACUC ETCR-2024-0329 · AVMA VCT #26005934.